MA3521 — Eigenvalue Systems

Name:

Problem 1 Find the GS, then use the IC to determine the constants

$$\mathbf{y}' = A\mathbf{y} \qquad A = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} \qquad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

1a. Eigenvalues and eigenvectors

$\det(A - \lambda I) =$ $= 0$

$\lambda_1 =$ $\mathbf{v}_1 =$

$\lambda_2 =$ $\mathbf{v}_2 =$

1b. General solution

$\mathbf{y}_c(t) =$

Write down $B\mathbf{c} = \mathbf{b}$ to solve for $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$B =$ $\mathbf{b} =$

$\mathbf{y}(t) =$ as $t \to \infty$ $\sim$

Problem 2 Find the GS, then use the IC to determine the constants

$$\mathbf{y}' = A\mathbf{y} \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \qquad \mathbf{y}(0) = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

2a. Eigenvalues and eigenvectors

$\det(A - \lambda I) =$ $= 0$

$\lambda_1 =$ $\mathbf{v}_1 =$

$\lambda_2 =$ $\mathbf{v}_2 =$

2b. General solution

$\mathbf{y}_c(t) =$

Write down $B\mathbf{c} = \mathbf{b}$ to solve for $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$B =$ $\mathbf{b} =$

$\mathbf{y}(t) =$ as $t \to \infty$ $\sim$